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Swap-it – Electric Drone Battery Swapping System

Project Specifications and Risk Assessment

Group 2017.032

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1 High Level Description of Project

1.1 Motivation

More applications for autonomous vehicles (AVs) are developing every year. Drones can monitor forests for fires [1], robots can discover new planets [2], and much more. The fascinating thing about AVs, whether it be a land robot or a drone, is that they can perform various tasks without human intervention. Thus, the dangers and responsibilities a human would previously be exposed to are shifted to the machine. This allows humans to explore places not previously possible. However, a significant issue facing AVs today is their limited use due to a poor battery lifespan. Current AVs either shutdown on the spot, or carefully time their task(s) to ensure that there is sufficient battery life to return to an operator or charging station where the battery is exchanged manually or set to charge for several hours. Pertaining to electric drones specifically, the best ones on the market only have an average flight time of 20 minutes [3].

1.2 Project Objective

The objective of this project is to design a fully automated battery swapping station for electric drones. Swap-it aims to reduce the current battery limitations imposed on electric drones by providing a fully automated station where the drone can go to swap its depleted battery with a charged one, allowing it to continue on with its duties in a matter of minutes. Swap-it can be used as a local or remote station, depending on the application.

1.3 Block Diagram

Swap-it is designed to detect the presence of an approaching drone and help guide its landing to a fixed location on the station. The swapping sequence will begin immediately after the drone has landed. Through a combination of digital control theory and precision mechanics, a microcontroller interfaces with a set of electric motors to precisely control the directional and rotational movements of the swapping system. The system extracts the drained battery from the drone and stores it for charging. It then retrieves a charged battery from the battery storage system and inserts it into the drone. The high-level system overview for Swap-it is shown in Figure 1 below.

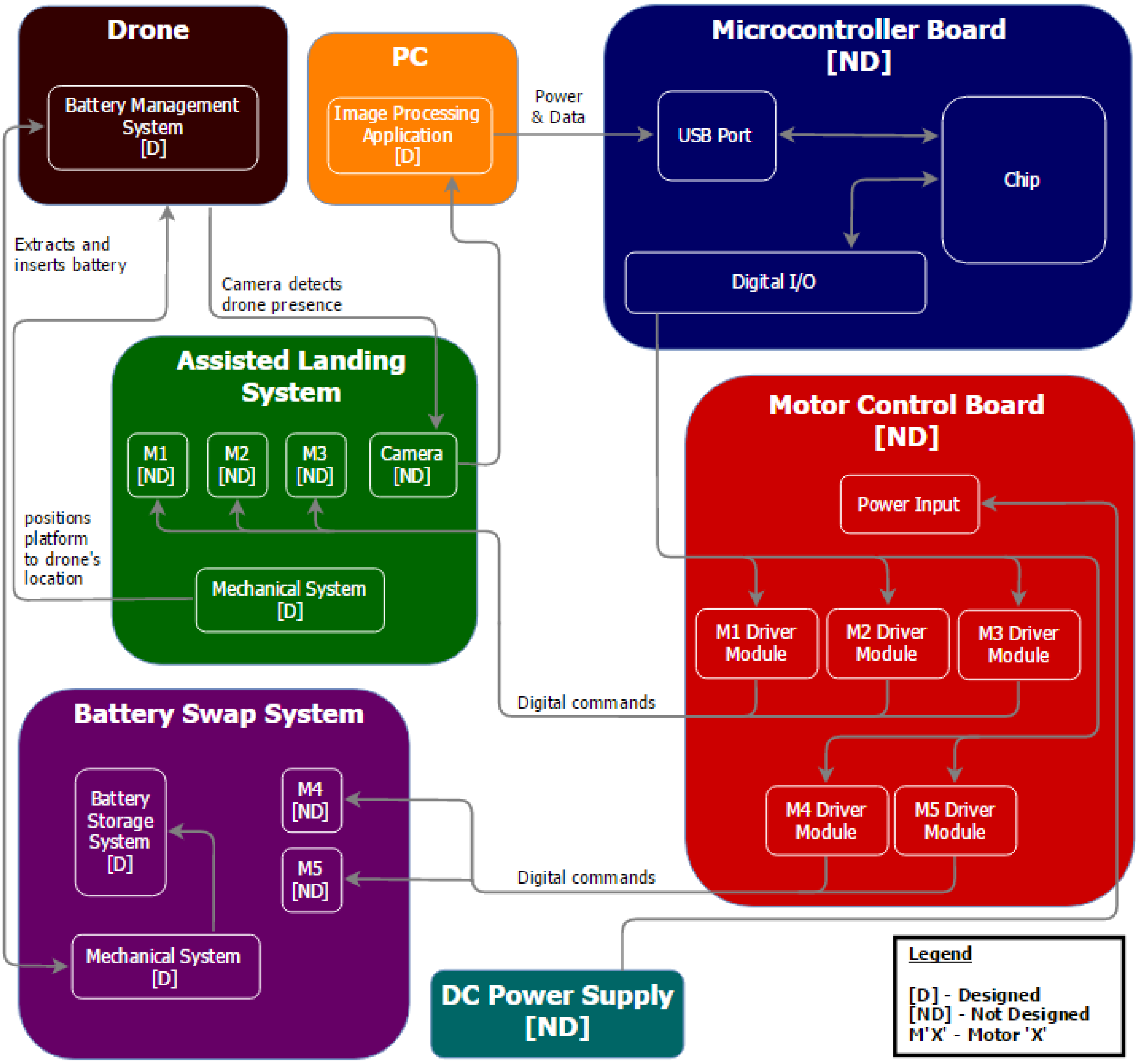

Figure 1. Block diagram showing the interconnectivities between subsystems.

1.3.1 Drone

The drone is equipped with a custom battery management system (BMS) located on its bottom so that the battery can be removed and reinserted by the swapping station. The BMS contains a hot swap system so that power is maintained during the battery swapping process.

1.3.2 PC

The role of the PC is to run the Image Processing Application, which is used to locate and track the drone during the landing process.

1.3.3 Assisted Landing System (ALS)

This subsystem consists of a moving platform with a sky-facing camera, driven by a set of three motors. The platform moves in the XY plane around the station to assist the drone's landing.

1.3.4 Battery Swap System (BSS)

The BSS is a mechanical process driven by a set of two motors to remove, store, and replace the drone's battery.

1.3.5 Microcontroller Board

The Microcontroller board provides the logic that processes the incoming data from the Image Processing Application to control the ALS and the BSS.

1.3.6 Motor Control Board (MCB)

The MCB consists of a set of motor drivers that interface the commands from the Microcontroller Board to the individual motors.

2 Project Specifications

2.1 Functional Specifications

This section details the functional specifications for this design. They are broken down into two categories: essential and non-essential. Table 1 lists the essential specifications that must be met in order to consider this design complete. Table 2 lists the non-essential specifications that add value to the project, but are not required.

Table 1. Essential Functional Specifications

Subsystem	Specification	Description
Assisted Landing System	Landing Precision	The moving platform must be able to align itself under the center of the drone within a 2cm tolerance.
Assisted Landing System	Incoming Drone Speed	The ALS shall be able to land a drone with an incoming speed of 0.2m/s.
Assisted Landing System	Drone Detection	The camera must be able to detect the incoming drone from a height of at least 1m above the station.
Image Processing Application	Tracking Latency	The application shall be able to start tracking the drone’s motion within 200ms from the time of detection.
Battery Swap System	Battery Swap	The drone’s battery must be replaced with a new battery after landing on the platform.
Battery Swap System	Battery Charging	Once removed from the drone, the drained battery shall be placed in a spot where it can be charged.
Drone	Drone Take Off	The drone must be able to continue flight after the battery has been swapped.
Battery Management System	Durability	The BMS must remain secure during regular drone operation until the drone has completely landed for batteries weighing up to 100 grams.
Motor Control Board	Precision	The motor drivers shall be able to control the rotation of the motor shafts within 3.6° of the desired angle.
Microcontroller Board	Communication and Control	The Microcontroller must be able to receive control information through a serial port and process the logic required to control at least 3 motors simultaneously.

Table 2. Non-Essential Functional Specifications

Subsystem	Specification	Description
Battery Management System	Hot Swap	The drone should not lose power during the battery swapping process.
Battery Swap System	Consecutive Battery Swap	The station shall be able perform consecutive battery swaps without human intervention.

2.2 Non-Functional Specifications

In addition to functional specifications, this project also contains non-functional specifications. Table 3 lists the essential non-functional specifications, and Table 4 lists the non-essential non-functional specifications.

Table 3. Essential Non-Functional Specifications

Subsystem	Specification	Description
Assisted Landing System	Size of Landing Area	The dimensions of the landing area shall be at least 50cm x 50cm.
Drone	Drone Payload	The payload added to the drone must be less than 1500g.
Power Supply	Supply Voltage/ Current	The system shall run off a 12V DC power supply and not consume more than 150W.

Table 4. Non-Essential Non-Functional Specifications

Subsystem	Specification	Description
Overall System	Total Size	The total size of the station shall not exceed 1m x 1m.
Overall System	Total Mass	The total mass of the station shall not exceed 15kg.

3 Risk Assessment

Every engineering project has potential risk factors associated with it. The risk factors for this project have been identified and are outlined in Table 5. Their probability of occurrence has been determined, as well as the potential impact they could have on the successful completion of this design. Prevention plans are listed for the high impact and high probability risk factors.

Table 5. Risk Factor Assessment

Risk Factor	Impact - Probability	Mitigation
Insufficient Time: A total battery swapping system for a drone involves many different subsystems, including landing assistance, battery swapping, and battery changing. The group has narrowed the scope of this project by choosing not to include the charging subsection, however the landing assistance and battery swapping sections will be challenging.	High - Medium	To reduce the likelihood of running out of time the group will stick to a detailed project timeline. This will ensure that team members don't spend too long on a single task. Weekly meetings with the project consultant are also scheduled to help keep the team on track. If time becomes an issue, team members have agreed to use their free time during the 4A co-op term to catch up.
Insufficient Mechanical Design Knowledge: This project relies heavily on mechanical design. All members of the group are part of the ECE program, so detailed mechanical design knowledge is lacking.	High - Medium	The team chose a consultant with experience in robotics and mechanical design to help minimize the impact of this void within the team. The group can also consult other mechanical and mechatronics professors for guidance on the landing assistance and battery swapping portions of the design.
Group Member Availability: Course work and job interviews may impact the availability of the group and jeopardize the project timeline.	Medium - Low	To reduce the probability of this risk occurring, group meetings were scheduled in the evenings so there would be a low chance of interviews conflicting. The group also uses Slack to keep everyone up to date on what is happening, even if someone has to miss a meeting.

References

[1] M. R. Roberts, "5 drone technologies for firefighting," 20 March 2014. [Online]. Available: http://www.firechief.com/2014/03/20/5-drone-technologies-firefighting/.

[2] E. Howell, "Opportunity Mars Rover Marks 12 Years on Red Planet," 26 January 2016. [Online]. Available: http://www.space.com/31735-opportunity-rover-12-years-mars.html.

[3] B. Carte, "RC Drones Review," 2016. [Online]. Available: http://rc-drones-review.toptenreviews.com/.